A Hybrid Matrix Completion + Liquid Neural Network Framework for Groundwater Level Imputation

A General Framework Coupling Spatial Correlation with Continuous-Time Dynamics for Sparse Well Networks

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1 The Problem

Groundwater monitoring networks worldwide contain **millions of measurements** but are plagued by **long temporal gaps** spanning months to years. Jasechko et al. (2024) found rapid declines in 30% of 1,693 aquifer systems globally, yet most wells lack continuous records needed for trend analysis.

>5yr	<50	0.3
Typical Max Gap	Median Obs/Well	Prior KGE at 2yr+

Prior methods (ELM, LSTM, GRU) treat each well independently in the temporal domain. Matrix completion has been applied only once to groundwater (Sharma et al., 2024) as a standalone method. No prior work couples spatial MC with temporal neural dynamics.

2 MC+LNN Pipeline

Stage 1: PCHIP — Fill gaps ≤ 24 months; Shape-preserving polynomial interpolation →
Stage 2a: ARCHI — Top-15 correlated donor wells via OLS regression; Trend-aware initialization →
Stage 2b: MC — SoftImpute SVD on composite matrix: donors + GLDAS + sin/cos →
Stage 2c: LNN — CFC reservoir dynamics; MC as input placeholder; Trained on observations only

Key insight: MC provides **spatial context** (cross-well correlations), LNN provides **temporal dynamics** (nonlinear continuous-time evolution). MC predictions drive the LNN reservoir, but readout trains only on real observations.

LNN CFC: $x(t+dt) = x \cdot e^{-leak \cdot dt} + (b / leak) \cdot (1 - e^{-leak \cdot dt})$ where $b = \tanh(W_{in} \cdot input + W_{res} \cdot x)$

3 What's New

First framework to jointly exploit:

Spatial Structure
Matrix Completion exploits correlation across the full 592-well network via low-rank SVD with GLDAS auxiliary rows and seasonal encoding

Temporal Dynamics
Liquid Neural Network with closed-form continuous-time cells (Hasani et al., 2022) captures nonlinear drought response, recharge lags, and seasonal patterns

vs. Prior approaches:

- **ELM** (Evans et al., 2020): Temporal only, no cross-well structure
- **LSTM/GRU** (Jeong et al., 2020; Wunsch et al., 2021): Each well independent
- **SoftImpute** (Sharma et al., 2024): Spatial only, no temporal model
- **MC+LNN (ours)**: Spatial + Temporal, coupled

4 Cross-Validation Results (Validated on 592 wells, Great Salt Lake Basin)

A. Random Missing Data (50 trials each)

% Removed	KGE	R^2	RMSE (ft)	MAE (ft)
5%	0.837 ± 0.085	0.771	2.53	1.98
10%	0.843 ± 0.077	0.770	2.70	2.06
20%	0.853 ± 0.051	0.787	2.64	1.98
30%	0.844 ± 0.048	0.770	2.79	2.03
40%	0.851 ± 0.030	0.784	2.75	2.03
50%	**0.847 ± 0.035**	**0.788**	**2.74**	**2.03**

KGE remains >0.84 even when 50% of data is removed.

B. Consecutive Year Gaps (20 trials each)

Gap Size	KGE	R^2	RMSE (ft)	MAE (ft)
1 year	0.783 ± 0.116	0.703	2.98	2.28
2 years	0.792 ± 0.081	0.721	2.97	2.29
3 years	0.802 ± 0.076	0.730	3.02	2.33
4 years	0.806 ± 0.055	0.738	2.96	2.15
5 years	**0.815 ± 0.063**	**0.744**	**2.91**	**2.20**

KGE >0.78 even for 5-year consecutive gaps.

5 PCHIP Matters

Replacing LNN-based small-gap fill with PCHIP in Stage 1 improves performance across the board:

+10%	+19%
Random KGE	Consec. KGE

PCHIP densification improves donor correlations for the MC stage, cascading through the pipeline.

6 Component Contributions

Each component's role (20 trials, same CV folds):

Method	KGE (30% random)	KGE (3-yr consec.)
MC only	0.783 ± 0.046	0.729 ± 0.067
LNN only	0.858 ± 0.032	0.849 ± 0.050
MC + LNN	**0.857 ± 0.037**	**0.807 ± 0.062**

MC captures spatial structure (cross-well correlations). **LNN** captures temporal dynamics via auxiliary forcings. The coupled framework provides **robustness across diverse well settings**.

7 General Framework, Validated on GSLB

Applicable to Any Well Network

- **Inputs:** Well locations, irregular observations, GLDAS auxiliary data (globally available)
- **No basin-specific tuning:** Donor selection, MC rank, and LNN hyperparameters are auto-optimized per well
- **Scales:** Tested on 592 wells; pipeline handles any network size
- **Handles heterogeneity:** WTE range 4,200-7,200 ft across GSLB; per-row normalization adapts to any range
- **Open source:** Runs in browser (TypeScript) or Python backend

Validation Basin: GSLB

Area: **93,000 km²**
Wells: **592** (2000-2023)
Months: **288**
Aux: **5** GLDAS soil moisture vars
Donors: **top 15** by correlation
MC ranks: **3, 5, 8, 10, 12** (auto)
LNN ensemble: **3 models**
Result: **100% completeness**

8 Conclusions

- **KGE > 0.84** under random missing data from 5-50%, with remarkable consistency across missing-data rates
- **KGE > 0.78 for 1-5 year consecutive gaps** -- the hardest imputation scenario where prior methods degrade to KGE < 0.3
- **MC + LNN are complementary:** MC captures spatial structure via cross-well correlations; LNN captures temporal dynamics via auxiliary forcings; the coupled framework provides robustness across diverse well settings
- **General-purpose:** No basin-specific tuning required. Uses globally available GLDAS auxiliary data and auto-optimized parameters
- **Scalable:** Validated on 592 wells; runs in browser or Python

Applicable to any groundwater monitoring network with sparse, irregular observations and multi-year gaps